



Ms-Excel

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Notes By:
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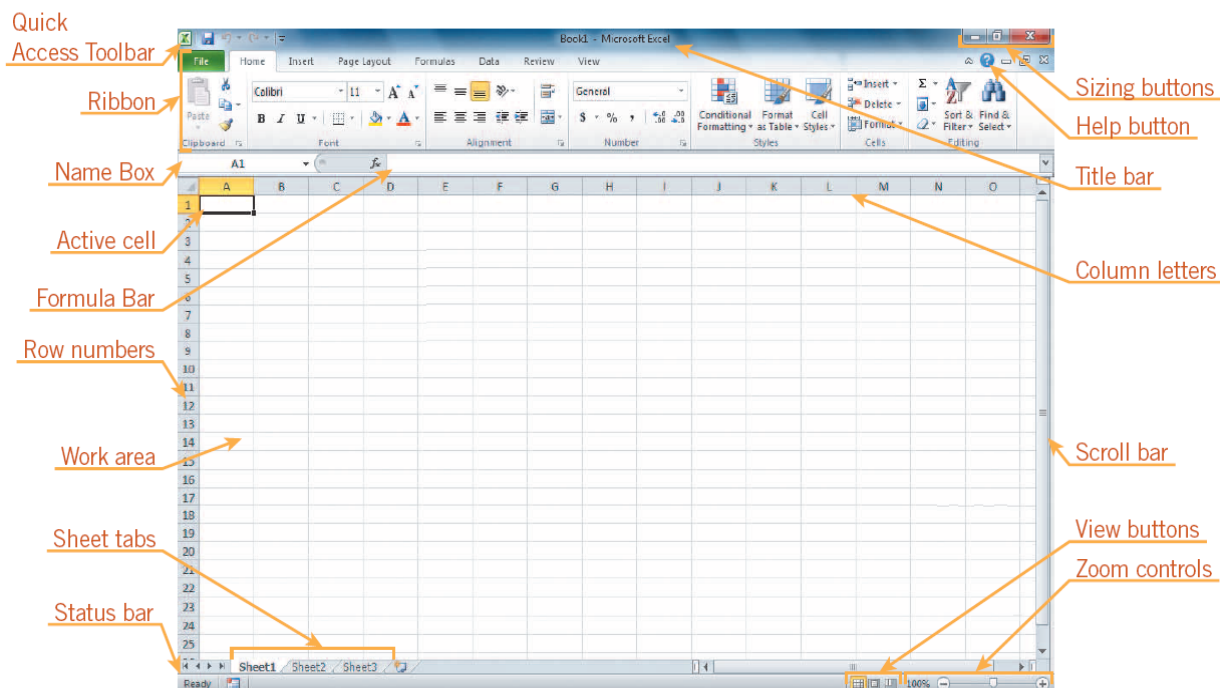
INTRODUCTION

MS-Excel is a Windows based Spreadsheet application package. It is quite useful in entering, editing, analysis and storing of data. Arithmetic operations with numerical data such as addition, subtraction, multiplication and division can also be done with Excel. You can sort the numbers/characters according to some given criteria (like ascending, descending etc.) and solve simple financial, mathematical and statistical formulas.

EXCEL FEATURES

There are a number of features that are available in Excel to make your task easier. Some of the main features are:

1. **AutoSum** - helps you to add the contents of a cluster of adjacent cells.
2. **List AutoFill** - automatically extends cell formatting when a new item is added to the end of a list.
3. **AutoFill** - feature allows you to quickly fill cells with repetitive or sequential data such as chronological dates or numbers, and repeated text. AutoFill can also be used to copy functions. You can also alter text and numbers with this feature.
4. **AutoShapes** toolbar will allow you to draw a number of geometrical shapes, arrows, flowchart elements, stars and more. With these shapes you can draw your own graphs.
5. **Drag and Drop** - feature will help you to reposition the data and text by simply dragging the data with the help of mouse.
6. **Charts** - features will help you in presenting a graphical representation of your data in the form of Pie, Bar, Line charts and more.
7. **PivotTable** - flips and sums data in seconds and allows you to perform data analysis and generating reports like periodic financial statements, statistical reports, etc. You can also analyse complex data relationships graphically.



Active Cell

- The active cell is recognized by its green outline. Data is always entered into the active cell. Different cells can be made active by clicking on them with the mouse or by using the arrow keys on the keyboard.

Add Sheet Icon

- Clicking on the *Add sheet icon* next to the *Sheet tab* at the bottom of the screen adds another worksheet. (You can also use two keyboard shortcuts to add a new worksheet: *Shift + F11* and *Alt + Shift + F1*.)

Cell

- Cells are the rectangular boxes located in central area of a worksheet.
- Data entered into a worksheet is stored in a cell. Each cell can hold only one piece of data at a time.
- A cell is the intersection point of a vertical column and a horizontal row.
- Each cell in the worksheet can be identified by a cell reference, which is a combination of letters and numbers such as A1, F456, or AA34.

Column Letters

- Columns* run vertically on a worksheet, and each one is identified by a letter in the column header.

Formula Bar

- Located above the worksheet, this area displays the contents of the active cell. The formula bar can also be used for entering or editing data and formulas.

Name Box

- Located next to the formula bar, the Name Box displays the cell reference or the name of the active cell.

Quick Access Toolbar

- The *Quick Access toolbar* allows you to add frequently used commands. Click on the down arrow at the end of the toolbar to display available options.

Ribbon

- The Ribbon is the strip of buttons and icons located above the worksheet. When clicked on, these buttons and icons activate the various features of the program. First introduced in Excel 2007, the ribbon replaced the menus and toolbars found in Excel 2003 and earlier versions.

Ribbon Tabs

- *Ribbon tabs* are part of the horizontal ribbon menu that contains links to various features of the program. Each tab—such as *Home*, *Page Layout*, and *Formulas*—contains a number of related features and options that are activated by clicking on the appropriate icon.

Opening Excel Sheet

You may have a shortcut to Word on your desktop, if so double click the icon and Word will open. If not follow the steps below:

1. Click on the Start button
2. Highlight Programs
3. Highlight Microsoft Office
4. Click on Microsoft Excel 2010

Create a New Workbook

1. Click the **File** tab and then click **New**.
2. Under **Available Templates**, double click **Blank Workbook** or Click **Create**.

Find and Apply Template

Excel 2010 allows you to apply built-in templates and to search from a variety of templates on Office.com. To find a template in Excel 2010, do the following:



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1. On the **File** tab, click **New**.
2. Under **Available Templates**, do one of the following:
 - a. To reuse a template that you've recently used, click **Recent Templates**, click the template that you want, and then click **Create**.
 - b. To use your own template that you already have installed, click **My Templates**, select the template that you want, and then click **OK**.
 - c. To find a template on Office.com, under **Office.com Templates**, click a template category, select the template that you want, and then click **Download** to download the template from Office.com to your computer.
3. Once you click on the template you like it will open on your screen as a new document.

Enter Data in a Worksheet

You can enter various kinds of data in a cell.

1. **Numbers:** Your numbers can be from the entire range of numeric values: whole numbers (example, 25), decimals (example, 25.67) and scientific notation (example, 0.2567E+2). Excel displays scientific notation automatically if you enter a number that is too long to be viewed in its entirety in a cell. You may also see number signs (#####) when a cell entry is too long. Widening the column that contains the cell with the above signs will allow you to read the number.
2. **Text:** First select the cell in which data has to be entered and type the text. Press ENTER key to finish your text entry. The text will be displayed in the active cell as well as in the Formula bar. If you have numbers to be treated as text use an apostrophe (') as the first character. You cannot do calculations with this kind of data entry.
3. **Date and Time:** When you enter dates and times, Excel converts these entries into serial numbers and kept as background information. However, the dates and times will be displayed to you on the worksheet in a format opted by you.
4. **Data in Series:** You can fill a range of cells either with the same value or with a series of values with the help of AutoFill.

EDITING DATA

Editing your Excel worksheet data is very easy. You can edit your data by any of the following ways:

1. Select the cell containing data to be edited. Press F2. Use Backspace key and erase the wrong entry. Retype the correct entry.
2. Select the cell and simply retype the correct entry.
3. If you want only to clear the contents of the cell, select the cell and press Delete key.

Select Cells or Ranges

In order to complete more advanced processes in Excel you need to be able to highlight or select cells, rows and columns. There are a variety of ways to do this, see the table below to understand the options.

Select All ctrl+A

Select On

Modifying Spreadsheets

In order to create an understandable and professional document you will need to make adjustments to the cells, rows, columns and text. Use the following processes to assist when creating a spreadsheet.

Cut, Copy, and Paste Data

You can use the Cut, Copy, and Paste commands in Microsoft Office Excel to move or copy entire cells or their contents. **NOTE:** Excel displays an animated moving border around cells that have been cut or copied. To cancel a moving border, press ESC.

Move/Copy Cells

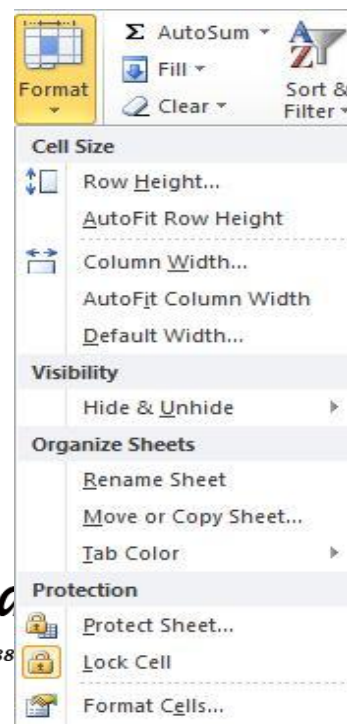
When you move or copy a cell, Excel moves or copies the entire cell, including formulas and their resulting values, cell formats, and comments.

1. Select the cells that you want to move or copy.
2. On the **Home** tab, in the **Clipboard** group, do one of the following:
 - a. To move cells, click Cut .
 - b. To copy cells, click Copy .
3. Click in the center of the cell you would like to Paste the information too.
4. On the Home tab, in the Clipboard group, click Paste .

NOTES: Excel replaces existing data in the paste area when you cut and paste cells to move them. When you copy cells, cell references are automatically adjusted. If the selected copy or paste area includes hidden cells, Excel also copies the hidden cells. You may need to temporarily unhide cells that you don't want to include when you copy information.

Move/Copy Cells with Mouse

1. Select the cells or a range of cells that you want to move or copy.



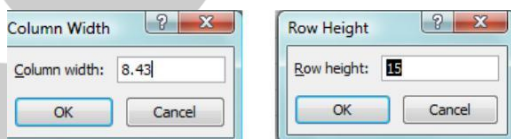
- To move a cell or range of cells, point to the border of the selection. When the pointer becomes a move pointer , drag the cell or range of cells to another location.

Column Width and Row Height

On a worksheet, you can specify a column width of 0 to 255 and a row height of 0 to 409. This value represents the number of characters that can be displayed in a cell that is formatted with the standard font. The default column width is 8.43 characters and the default row height is 12.75 points. If a column/row has a width of 0, it is hidden.

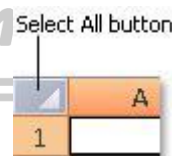
Set Column/Row Width/Height

- Select the column(s) or row(s) that you want to change.
- On the **Home** tab, in the **Cells** group, click **Format**.
- Under **Cell Size**, click **Column Width** or **Row Height**.
- A **Column Width** or **Row Height** box will appear.
- In the **Column Width** or **Row Height** box, type the value that you want your column or row to be.



Automatically Fit Column/Row Contents

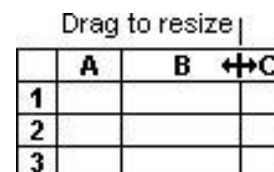
- Click the **Select All** button
- Double-click any boundary between two column/row headings.
- All Columns/Rows in the entire worksheet will be changed to the new size



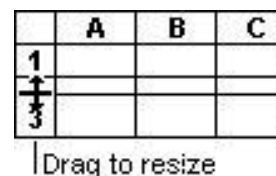
NOTE: At times, a cell might display #####. This can occur when the cell contains a number or a date that exceeds the width of the cell so it cannot display all the characters that its format requires. To see the entire contents of the cell with its current format, you must increase the width of the column.

Set Column/Row Width/Height with Mouse

- To change the width of one column/row
 - Place your cursor on the line between two rows or columns.
 - A symbol that looks like a lower case t with arrows on the horizontal line will appear
 - Drag the boundary on the right side of the column/row heading until the column/row is the width that you want.
- To change the width of multiple columns/rows



1. Select the columns/rows that you want to change
2. Drag a boundary to the right of a selected column/row heading.
3. All selected columns/rows will become a different size.



- To change the width of columns/rows to fit the contents in the cells

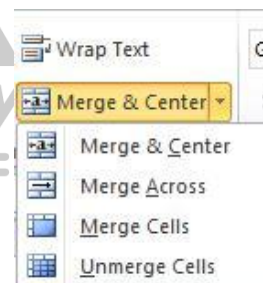
1. Select the column(s) or row(s) that you want to change
2. Double-click the boundary to the right of a selected column/row heading.
3. The Column/Row will automatically be size to the length/height of the longest/tallest text.

Merge or Split Cells

When you merge two or more adjacent horizontal or vertical cells the cells become one larger cell that is displayed across multiple columns or rows. When you merge multiple cells, the contents of only one cell appear in the merged cell.

Merge and Center Cells

1. Select two or more adjacent cells that you want to merge.
2. On the **Home** tab, in the **Alignment** group, click **Merge and Center**.
3. The cells will be merged in a row or column, and the cell contents will be centered in the merged cell.



Merge Cells

To merge cells only, click the arrow next to **Merge and Center**, and then click **Merge Across** or **Merge Cells**.

Split Cells

1. Select the merged cell you want to split
2. To split the merged cell, click **Merge and Center** . The cells will split and the contents of the merged cell will appear in the upper-left cell of the range of split cells.

Automatically Fill Data

To quickly fill in several types of data series, you can select cells and drag the fill handle . To use the fill handle, you select the cells that you want to use as a basis for filling additional cells, and then drag the fill handle across or down the cells that you want to fill.

1. Select the cell that contains the formula that you want to be brought to other cells.

2. Move your cursor to the small black square in the lower-right corner of a selected cell also known as the fill handle. Your pointer will change to a small black cross. 
3. Click and hold your mouse then drag the fill handle across the cells, horizontally to the right or vertically down, that you want to fill.
4. The cells you want filled will have a gray looking border around them. Once you fill all of the cells let go of your mouse and your cells will be populated.

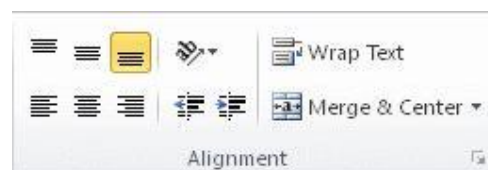
Formatting Spreadsheets

To further enhance your spreadsheet you can format a number of elements such as text, numbers, coloring, and table styles. Spreadsheets can become professional documents used for company meetings or can even be published.

Wrap Text

You can display multiple lines of text inside a cell by wrapping the text. Wrapping text in a cell does not affect other cells.

1. Click the cell in which you want to wrap the text.
2. On the **Home** tab, in the **Alignment** group, click **Wrap Text**.
3. The text in your cell will be wrapped.

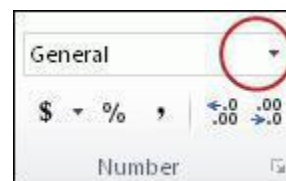


NOTE: If the text is a long word, the characters won't wrap (the word won't be split); instead, you can widen the column or decrease the font size to see all the text. If all the text is not visible after you wrap the text, you might have to adjust the height of the row. On the **Home** tab, in the **Cells** group, click **Format**, and then under **Cell Size** click **AutoFit Row**

Format Numbers

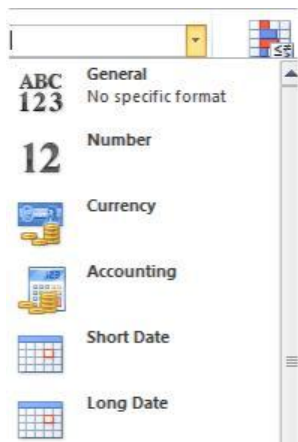
In Excel, the format of a cell is separate from the data that is stored in the cell. This display difference can have a significant effect when the data is numeric. For example, numbers in cells will default as rounded numbers, date and time may not appear as anticipated. After you type numbers in a cell, you can change the format in which they are displayed to ensure the numbers in your spreadsheet are displayed as you intended.

1. Click the cell(s) that contains the numbers that you want to format.
2. On the **Home** tab, in the **Number** group, click the arrow next to the



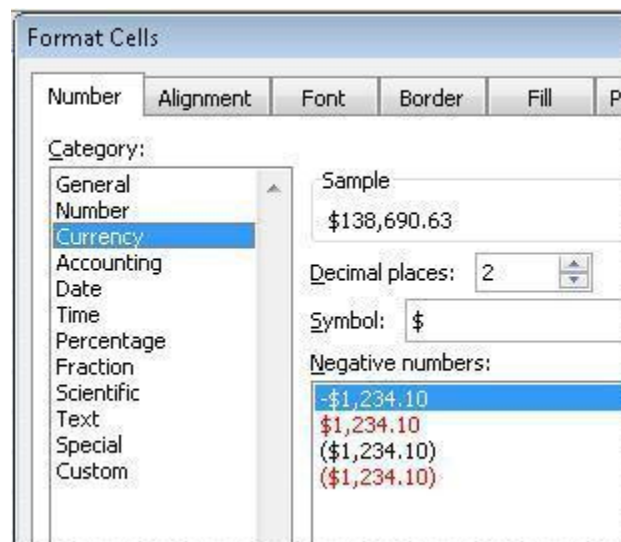
that

Number Format box, and then click the format you want.



If you are unable to format numbers in the detail you would like that you can click on the **More Number Formats** at the bottom of the **Number Format** drop down list.

1. In the Category list, click the format that you want to use, and then adjust settings to the right of the Format Cells dialog box. For example, if you're using the Currency format, you can select a different currency symbol, show more or fewer decimal places, or change the way negative numbers are displayed.



Cell Borders

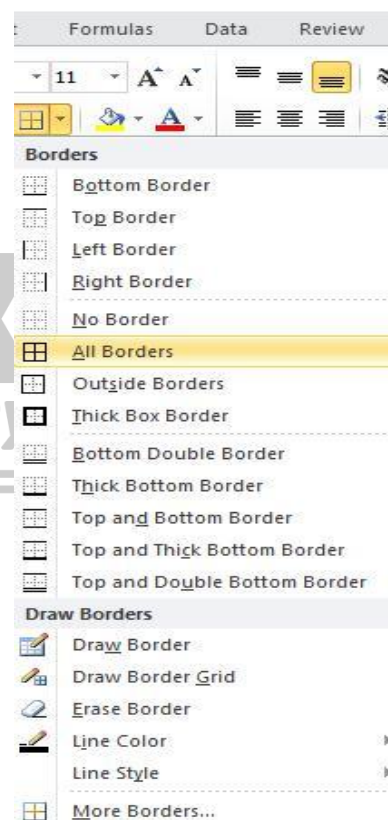
By using predefined border styles, you can quickly add a border around cells or ranges of cells. If predefined cell borders do not meet your needs, you can create a custom border.

NOTE: Cell borders that you apply appear on printed pages. If you do not use cell borders but want worksheet gridline borders for all cells to be visible on printed pages, you can display the gridlines.

Apply Cell Borders

1. On a worksheet, select the cell or range of cells that you want to add a border to, change the border style on, or remove a border from.
2. Go to the **Home** tab, in the **Font** group
3. Click the arrow next to **Borders**
4. Click on the border style you would like
5. The border will be applied to the cell or cell range

NOTE: To apply a custom border style, click **More Borders**. In the **Format Cells** dialog box, on the **Border** tab, under **Line** and **Color**, click the line style and color that you want.



Remove Cell Borders

1. Go to the **Home** tab, in the **Font** group
2. Click the arrow next to **Borders** 
3. Click **No Border** .

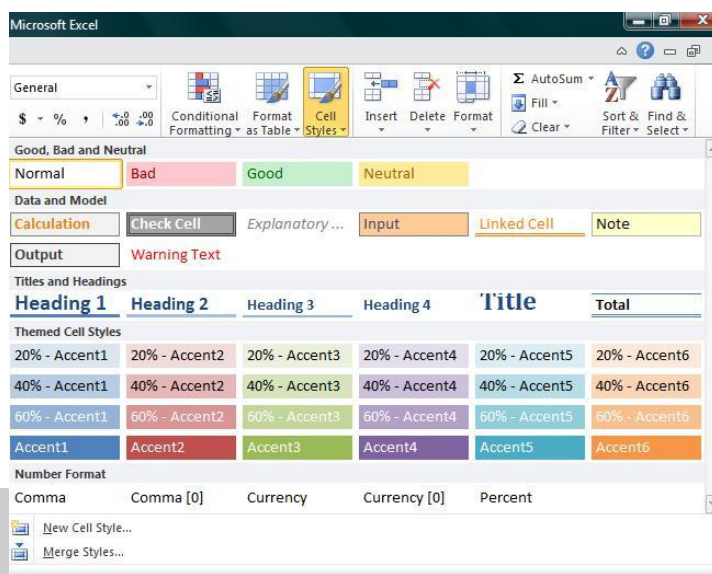
NOTES: The **Borders** button displays the most recently used border style. You can click the **Borders** button (not the arrow) to apply that style.

Cell Styles

You can create a cell style that includes a custom border, colors and accounting formatting.

1. On the **Home** tab, in the **Styles** group, click **Cell Styles**.
2. Select the different cell style option you would like applied to your spreadsheet.

NOTE: If you would like to apply a cell fill and a cell border, select the cell fill color first the ensure both formats are applied.

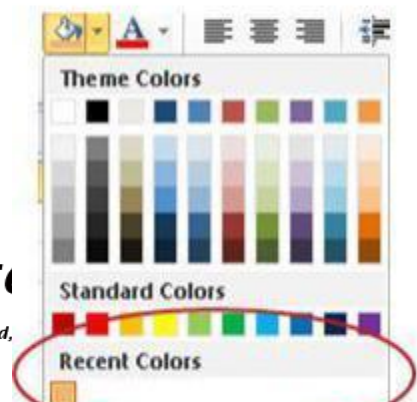


Cell and Text Coloring

You can also modify a variety of cell and text colors manually.

Cell Fill

1. Select the cells that you want to apply or remove a fill color from.
2. Go to the **Home** tab, in the **Font** group and select one of the following options:
 - a. To fill cells with a solid color, click the arrow next to **Fill Color** , and then under **Theme Colors** or **Standard Colors**, click the color that you want.
 - b. To fill cells with a custom color, click the arrow next to **Fill Color** , click **More Colors**, and then in the **Colors** dialog box select the color that you want.
 - c. To apply the most recently selected color, click **Fill Color** .



NOTE: Microsoft Excel saves your 10 most recently selected custom colors. To quickly apply one of these colors, click the arrow next to **Fill Color** , and then click the color that you want under **Recent Colors**.

Remove Cell Fill

1. Select the cells that contain a fill color or fill pattern.
2. On the **Home** tab, in the **Font** group, click the arrow next to **Fill Color**, and then click **No Fill**.

Text Color

1. Select the cell, range of cells, text, or characters that you want to format with a different text color.
2. On the **Home** tab, in the **Font** group and select one of the following options:
 - a. To apply the most recently selected text color, click **Font Color** .
 - b. To change the text color, click the arrow next to **Font Color** , and then under **Theme Colors** or **Standard Colors**, click the color that you want to use.

Bold, Underline and Italics Text

1. Select the cell, range of cells, or text.
2. Go to the **Home** tab, in the **Font** group
3. Click on the Bold (**B**) Italics (*I*) or Underline (U) commands.
4. The selected command will be applied.



Customize Worksheet Tab

1. On the **Sheet tab** bar, right-click the sheet tab that you want to customize
2. Click **Rename** to rename the sheet or **Tab Color** to select a tab color.
3. Type in the name or select a color you would like for your spreadsheet.
4. The information will be added to the tab at the bottom of the spreadsheet.



Formulas in Excel

Formulas are equations that perform calculations on values in your worksheet. A formula always starts with an equal sign (=). An example of a simple is **=5+2*3** that multiplies two numbers and then adds a number to the result. Microsoft Office Excel follows the standard order of mathematical operations. In the preceding example, the multiplication operation (2*3) is performed first, and then 5 is added to its result.

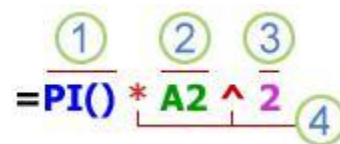
You can also create a formula by using a function which is a prewritten formula that takes a value, performs an operation and returns a value. For example, the formulas **=SUM(A1:A2)** and **SUM(A1,A2)** both use the **SUM** function to add the values in cells A1 and A2.

Depending on the type of formula that you create, a formula can contain any or all of the following parts. **1 Functions** A function, such as **PI()** or **SUM()**, starts with an equal sign (=).

2 Cell references You can refer to data in worksheet cells by including cell references in the formula. For example, the cell reference **A2** returns the value of that cell or uses that value in the calculation.

3 Constants You can also enter constants, such as numbers (such as **2**) or text values, directly into a formula.

4 Operators Operators are the symbols that are used to specify the type of calculation that you want the formula to perform.

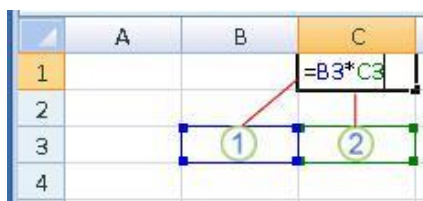


Create a Simple Formulas

1. Click the cell in which you want to enter the formula.
2. Type = (equal sign).
3. Enter the formula by typing the constants and operators that you want to use in the calculation.
4. Press ENTER.

Create a Formula with Cell References

EXAMPLE	WHAT IT DOES
FORMULA	
=5+2	Adds 5 and 2
=5-2	Subtracts 2 from 5
=5/2	Divides 5 by 2
=5*2	Multiplies 5 times 2
=5^2	Raises 5 to the 2nd power



1 The first cell reference is B3, the color is blue, and the cell range has a blue border with square corners.

2 The second cell reference is C3, the color is green, and the cell range has a green border with square corners. To create your formula:

1. Click the cell in which you want to enter the formula.
2. In the **formula bar**, at the top of the Excel window that you use, , type = (equal sign).
3. Click on the 1st cell you want in the formula.

4. Enter an Operator such

as +, or *.

EXAMPLE WHAT IT DOES

FORMULA

5. Click on the next cell you want in the formula.

=A1+A2 Adds the values in cells A1 and A2

=A1-A2 Subtracts the value in cell A2 from the value in A1

Continue steps 3 – 5 until

=A1/A2 Divides the value in cell A1 by the value in A2

the formula is complete

=A1*A2 Multiplies the value in cell A1 times the value in A2

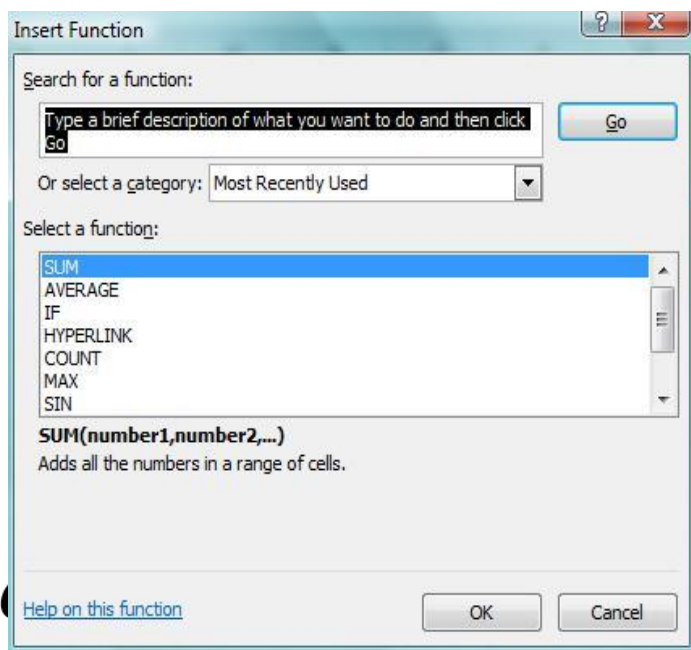
6. Hit the **ENTER** key on

your keyboard.

=A1^A2 Raises the value in cell A1 to the exponential value specified in A2

Create a Formula with Function

1. Click the cell in which you want to enter the formula.
2. Click **Insert Function**  on the formula bar . Excel inserts the equal sign (=) for you.
3. Select the function that you want to use.
NOTE: If you're not sure which function to use, type a question that describes what you want to do in the **Search for a function** box (for example, "add numbers" returns the **SUM**

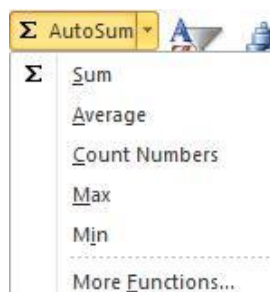


function), or browse from the categories in the

Or Select a category box.

4. Enter the arguments.
5. After you complete the formula, press ENTER.

Use Auto Sum



To summarize values quickly, you can also use **AutoSum**.

1. Select the cell where you would like your formulas solution to appear.
2. Go to the **Home** tab, in the **Editing** group,
3. Click **AutoSum**, to sum your numbers or click the arrow next to **AutoSum** to select a function that you want to apply.

Delete a Formula

When you delete a formula, the resulting values of the formula is also deleted. However, you can instead remove the formula only and leave the resulting value of the formula displayed in the cell.

- To delete formulas along with their resulting values, do the following:

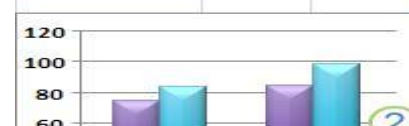
1. Select the cell or range of cells that contains the formula.
2. Press DELETE.

- To delete formulas without removing their resulting values, do the following:

1. Select the cell or range of cells that contains the formula.
2. On the **Home** tab, in the **Clipboard** group, click **Copy** .
3. On the **Home** tab, in the **Clipboard** group, click the arrow below **Paste** , and then click **Paste Values**.



	QTR1	QTR2
Projected	75	85
Actual	84	99



Charts in Excel

Microsoft Excel no longer provides the chart wizard. Instead, you can create a basic chart by clicking the chart type that you want on the Insert tab in the Charts group. Charts are used to display series of numeric data in a graphical format



to make it easier to understand large quantities of data and the relationship between different series of data.

To create a chart in Excel, you start by entering the numeric data for the chart on a worksheet. Then you can plot that data into a chart by selecting the chart type that you want to use on the Insert tab, in the Charts group.

1. Worksheet data
2. Chart created from worksheet data

Getting to know the elements of a chart

A chart has many elements. Some of these elements are displayed by default, others can be added as needed. You can change the display of the chart elements by moving them to other locations in the chart, resizing them, or by changing the format. You can also remove chart elements that you do not want to display.

- 1 The **chart area** is the entire chart and all its elements
- 2 The **plot area** is the area of the chart bounded by the axes.
- 3 The **data points** are individual values plotted in a chart represented by bars, columns, lines, or pies.
- 4 The **horizontal** (category) and **vertical** (value) **axis** along which the data is plotted in the chart.
- 5 The **legend** identifies the patterns or colors that are assigned to the data series or categories in the chart.
- 6 A **chart and axis title** are descriptive text that for the axis or chart.
- 7 A **data label** provides additional information about a data marker that you can use to identify the details of a data point in a data series.

Create a Chart

1. On the worksheet, arrange the data that you want to plot in a chart. The data can be arranged in rows or columns — Excel automatically determines the best way to plot the data in the chart.
2. Select the cells that contain the data that you want to use for the chart.

NOTE: If the cells that you want to plot in a chart are not in a continuous range, you can select nonadjacent cells or ranges as long as the selection forms a rectangle. You can also hide the rows or columns that you do not want to plot in the chart.



3. Go to the **Insert** tab, in the **Charts**
4. Click the chart type, and then click a chart subtype from the drop menu that will appear.
5. Click anywhere in the embedded chart to activate it. When you click on the chart, **Chart Tools** will be displayed which includes the **Design**, **Layout**, and **Format** tabs.
6. The chart will automatically be embedded in the worksheet. A chart name will automatically be assigned.

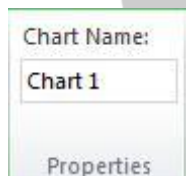
Move Chart to New Sheet



1. On the **Design** tab, in the **Location** group, click **Move Chart**.
2. Under **Choose where you want the chart to be placed**, click on the **New sheet** bubble
3. Type a chart name in the **New sheet** box.

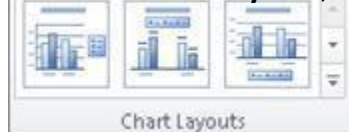
Change Chart Name

1. Click the chart.
2. On the **Layout** tab, in the **Properties** group, click the **Chart Name** text box.
3. Type a new chart name.
4. Press ENTER.



Change Chart Layout

1. Click anywhere in the chart.
2. Go to the **Chart Tools**, the **Design** group
3. In the **Chart Layouts**, click the chart layout that you want to use. To see all available layouts, click



More .

Change Chart Style

1. Click anywhere in the chart.
2. On the **Design** tab, in the **Chart Styles** group, click the chart style that you want to use. To see all predefined chart styles, click More .



Chart or Axis Titles

To make a chart easier to understand, you can add titles, such as chart and axis titles.

To add a chart title:

1. Click anywhere in the chart.
2. On the **Layout** tab, in the **Labels** group, click **Chart Title**.
3. Click **Centered Overlay Title** or **Above Chart**.
4. In the **Chart Title** text box that appears in the chart, type the text that you want.
5. To remove a chart title, click **Chart Title**, and then click **None**.

NOTE: You can also use the formatting buttons on the ribbon (**Home** tab, **Font** group). To format the whole title, you can right-click it, click **Format Chart Title**, and then select the formatting options that you want.

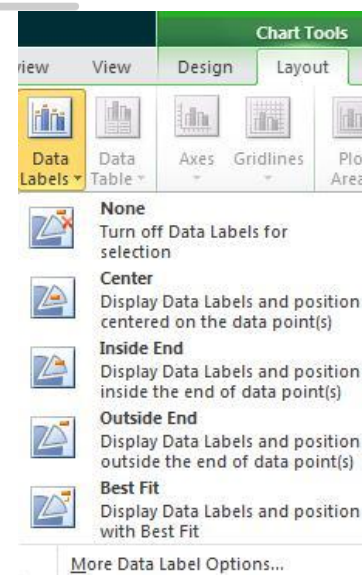


To add axis titles:

1. Click anywhere in the chart.
2. On the **Layout** tab, in the **Labels** group, click **Axis Titles**.
3. Do one or more of the following:
 - a. To add a title to a primary horizontal (category) axis, click **Primary Horizontal Axis Title**, and then click the option that you want.
 - b. To add a title to primary vertical (value) axis, click **Primary Vertical Axis Title**, and then click the option that you want.
4. In the Axis Title text box that appears in the chart, type the text that you want.
5. To remove an axis title, click **Axis Title**, click the type of axis title to remove, and then click **None**.

Data Labels

1. On a chart, do one of the following:
 - a. Click on the **chart area** to add a data label to all data points of all data series
 - b. Click in the **data series** to add a data label to all data points of a data series
 - c. Click on a specific **data point** to add a data label to a single data point in a data series
2. On the **Layout** tab, in the **Labels** group, click **Data Labels**, and then click the display option that you want.
3. Text boxes will appear in the area of your chart based on your selection.
4. Click on the text box to modify the text.
5. To remove data labels, click **Data Labels**, and then click **None**.



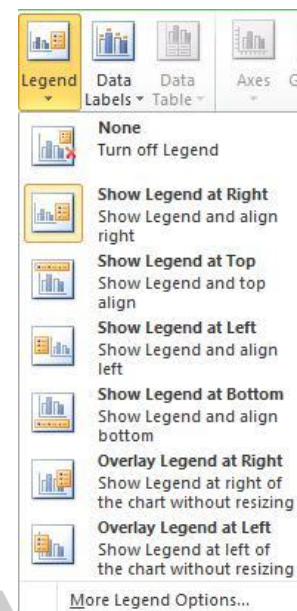
NOTE: Depending on the chart type that you used, different data label options will be available.

Legend

When you create a chart, the legend appears, but you can hide the legend or change its location after you create the chart.

1. Click the chart in which you want to show or hide a legend.
2. On the **Layout** tab, in the **Labels** group, click **Legend**.
3. Do one of the following:
 - a. To hide the legend, click **None**.
 - b. To display a legend, click the display option that you want.
 - c. For additional options, click More Legend Options, and then select the display option that you want.

NOTE: To quickly remove a legend or a legend entry from a chart, you can select it, and then press DELETE. You can also right-click the legend or a legend entry, and then click Delete.



Move or Resize Chart

You can move a chart to any location on a worksheet or to a new or existing worksheet. You can also change the size of the chart for a better fit. To move a chart, drag it to the location that you want.

To resize a chart, click on one of the edges and drag towards the center.

Advanced Spreadsheet Modification

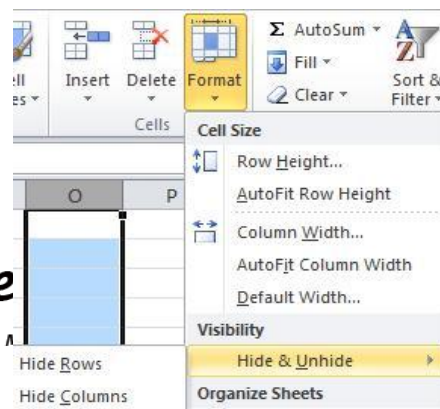
Once you have created a basic spreadsheet there are numerous things you can do to make working with you data easier. Some of these elements are hiding, freezing and splitting rows. You can also sort and filter data, these features are quite helpful when working with a large amount of data.

Hide or Display Rows and Columns

You can hide a row or column by using the **Hide** command or when you change its row height or column width to 0 (zero). You can display either again by using the **Unhide** command. You can either unhide specific rows and columns, or you can unhide all hidden rows and columns at the same time. The first row or column of the worksheet is tricky to unhide, but it can be done.

Hide Rows or Columns

1. Select the rows or columns that you want to hide.



2. On the **Home** tab, in the **Cells** group, click **Format**.
3. Under **Visibility**, point to **Hide & Unhide**, and then click **Hide Rows** or **Hide Columns**.

NOTE: You can also right-click a row or column (or a selection of multiple rows or columns), and then click **Hide**.

Unhide Rows or Columns

1. Select the rows, columns or entire sheet to unhide.
2. On the **Home** tab, in the **Cells** group, click **Format**.
3. Under **Visibility**, point to **Hide & Unhide**, and then click **Unhide Rows** or **Unhide Columns**.

TIP You can also right-click the selection of visible rows and columns surrounding the hidden rows and columns, and then click **Unhide**.

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Freezing/Splitting Rows and Columns

To keep an area of a worksheet visible while you scroll to another area of the worksheet, you can either lock specific rows or columns in one area by freezing panes.

Freezing vs. splitting

When you **freeze** panes, Microsoft Excel keeps specific rows or columns visible when you scroll in the worksheet. For example, if the first row in your spreadsheet contains labels, you might freeze that row to make sure that the column labels remain visible as you scroll down in your spreadsheet. A solid line indicates that the row is frozen to keep column labels in place when you scroll.

	A	B	C
1	City	Date	Books Sold
7	Boise	Mar	84
8	Denver	Mar	38
9	Chicago	Feb	60
10	Dallas	Feb	72
11	Santa Fe	Mar	154

When you **split** panes, Excel creates either two or four separate worksheet areas that you can scroll within, while rows or columns in the non-scrolled area remain visible. This worksheet has been split into four areas. Notice that each area contains a separate view of the same data. Splitting panes is useful when you want to see different parts of a large spreadsheet at the same time.

NOTE: You cannot split panes and freeze panes at the same time.

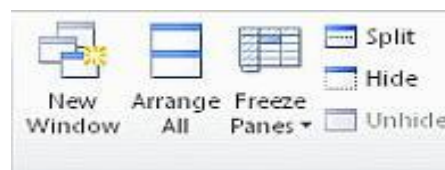
	A	B	C		A	B	C
1	ID	Country	Sales Region		ID	Country	Sales Region
2	1	Denmark	North		1	Denmark	North
3	2	Finland	North		2	Finland	North
4	3	France	Mid		3	France	Mid
5	4	Germany	Mid		4	Germany	Mid
1	ID	Country	Sales Region		ID	Country	Sales Region
2	1	Denmark	North		1	Denmark	North
3	2	Finland	North		2	Finland	North
4	3	France	Mid		3	France	Mid
5	4	Germany	Mid		4	Germany	Mid
6	5	Italy	South		5	Italy	South

Freeze Panes

1. On the worksheet, select the row or column that you want to keep visible when you scroll.

2. On the **View** tab, in the **Window** group, click the arrow below **Freeze Panes**.
3. Then do one of the following:

- To lock one row only, click **Freeze Top Row**.
- To lock one column only, click **Freeze First Column**.
- To lock more than one row or column, or to lock both rows and columns at the same time, click **Freeze Panes**.



NOTE: You can freeze rows at the top and columns on the left side of the worksheet only. You cannot freeze rows and columns in the middle of the worksheet.

Unfreeze panes

1. On the **View** tab, in the **Window** group, click the arrow below **Freeze Panes**.
2. Click **Unfreeze Panes**.

Split Panes



1. To split panes, point to the split box at the top of the vertical scroll bar or at the right end of the horizontal scroll bar.
2. When the pointer changes to a split pointer $\frac{\updownarrow}{\leftarrow\rightarrow}$ or $\frac{\leftarrow\rightarrow}{\updownarrow}$, drag the split box down or to the left to the position that you want.

3. To remove the split, double-click any part of the split bar that divides the panes.

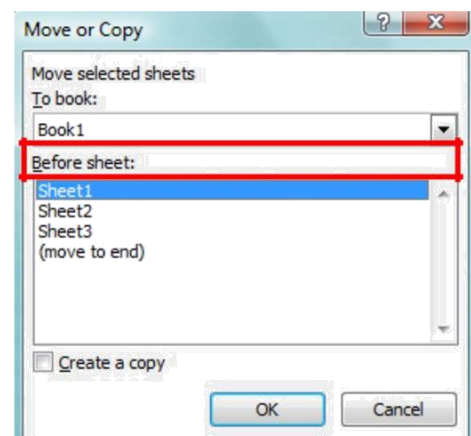
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Moving or Copying Worksheets

Sometimes you may need to copy an entire worksheet instead of copying and pasting the data which may or may not paste properly, you can use the steps below to achieve a much better result.

Move or Copy Worksheets

1. Select the worksheets that you want to move or copy.
2. On the **Home** tab, in the **Cells** group, click **Format**, and then under **Organize Sheets**, click **Move or Copy Sheet**. You can also right-click a selected sheet tab, and then click **Move or Copy**.
3. A **Move or Copy** dialog box will appear
4. To move a sheet, in the **Before sheet** list:
 - Click the sheet that you want to insert the moved or copied sheets directly in front of.
 - Click **move to end** to insert the moved or copied sheets after the last sheet in the workbook and before the **Insert Worksheet** tab.
5. To copy the sheets, in the **Move or Copy** dialog box, select the **Create a copy** check box.



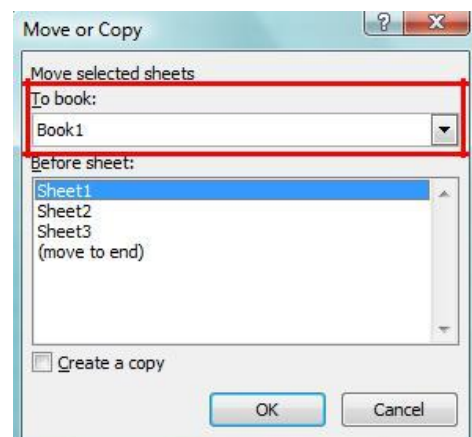
NOTE: When you create a copy of the worksheet, the

worksheet is duplicated in the workbook, and the sheet name indicates that it is a copy — for example, the first copy that you make of Sheet1 is named Sheet1 (2).

NOTE: To move sheets in the current workbook, you can drag the selected sheets along the row of sheet tabs. To copy the sheets, hold down CTRL, and then drag the sheets; release the mouse button before you release the CTRL key.

Move or Copy to a Different Workbook

1. In the workbook that contains the sheets that you want to move or copy, select the sheets.
2. On the **Home** tab, in the **Cells** group, click **Format**, and then under **Organize Sheets**, click **Move or Copy Sheet**.
3. In the **Move or Copy** dialog box, click the drop down list in the **To book** box, then:
 - Click the workbook to which you want to move or copy the selected sheets.
 - Click **new book** to move or copy the selected sheets to a new workbook.
6. To move a sheet, in the **Before sheet** list:
 - Click the sheet that you want to insert the moved or copied sheets directly in front of.
 - Click **move to end** to insert the moved or copied sheets after the last sheet in the workbook and before the **Insert Worksheet** tab.
7. To copy the sheets, in the **Move or Copy** dialog box, select the **Create a copy** check box.



Sorting Data

Sorting data is an integral part of data analysis. You might want to arrange a list of names in alphabetical order, compile a list of product inventory levels from highest to lowest, or order rows by colors or icons. Sorting data helps you quickly visualize and understand your data better, organize and find the data that you want, and ultimately make more effective decisions.

Sort Data in Single Column

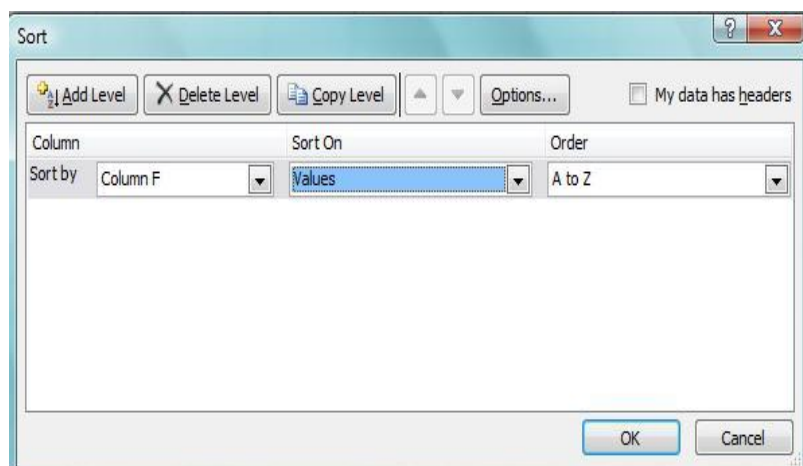
1. Select a column of data in a range of cells
2. On the **Data** tab, in the **Sort & Filter** group, do one of the following:
 - To sort in ascending or smallest to largest order, click **Sort A to Z**.
 - To sort in descending or largest to smallest order, click **Sort Z to A**.
3. To reapply a sort after you change the data, click a cell in the range or table and then, on the **Data** tab, in the **Sort & Filter** group, click **Reapply**.



Sort Data in Multiple Columns or Rows

You may want to sort by more than one column or row when you have data that you want to group by the same value in one column or row, and then sort another column or row within that group of equal values.

1. Select a range of cells with two or more columns of data.
2. On the **Data** tab, in the **Sort & Filter** group, click **Sort**.
3. The **Sort** dialog box will appear.
4. Under **Column**, in the **Sort by** box, select the first column that you want to sort.
5. Under **Sort On**, select the type of sort.
 - To sort by text, number, or date and time, select **Values**.
 - To sort by format, select **Cell Color**, **Font Color**, or **Cell Icon**.
6. Under **Order**, select how you want to sort.
 - For text values, select **A to Z** or **Z to A**.
 - For number values, select **Smallest to Largest** or **Largest to Smallest**.
 - For date or time values, select **Oldest to Newest** or **Newest to Oldest**.
 - To sort based on a custom list, select **Custom List**.
7. To add another column to sort by, click **Add Level**, and then repeat steps four through six.
8. To copy a column to sort by, select the entry and then click **Copy Level**.
9. To delete a column to sort by, select the entry and then click **Delete Level**.
10. To change the order in which the columns are sorted, select an entry and then click the **Up** or **Down** arrow to change the order.
11. To reapply a sort after you change the data, click a cell in the range or table and then, on the **Data** tab, in the **Sort & Filter** group, click **Reapply**.

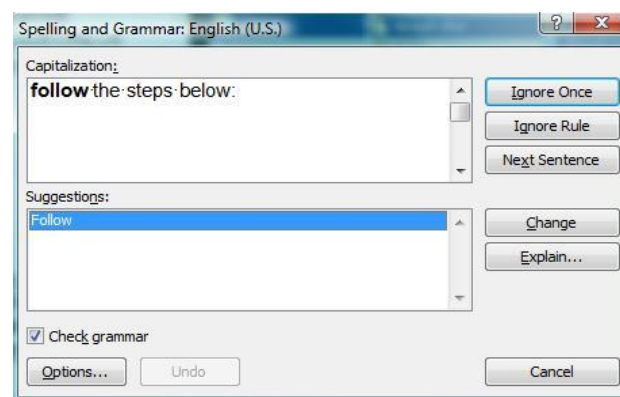


Finalizing a Spreadsheet

To complete your spreadsheet there are a few steps to take to ensure your document is finalized. Using the "Spell Check" Feature

Excel does not have the same spell check feature as Word and PowerPoint. To complete a Spelling and Grammar check, you need to use the Spelling and Grammar feature.

1. Click on the **Review** tab
2. Click on the **Spelling & Grammar** command (a blue check mark with ABC above it).



3. A **Spelling and Grammar** box will appear, correct any Spelling or Grammar issue with the help of the box.

Save a Spreadsheet

To save a document in the format used by Excel 2010 and Excel 2007, do the following:

1. Click the **File** tab.
2. Click **Save As**.
3. In the **File name** box, enter a name for your document.
4. Click **Save**.

To save a document so that it is compatible with Excel 2003 or earlier, do the following:

1. Click the **File** tab.
2. Click **Save As**.
3. In the **Save as type** list, click **Excel 97-2003 Document**. This changes the file format to .xls.
4. In the **File name** box, type a name for the document.
5. Click **Save**.

Print Preview

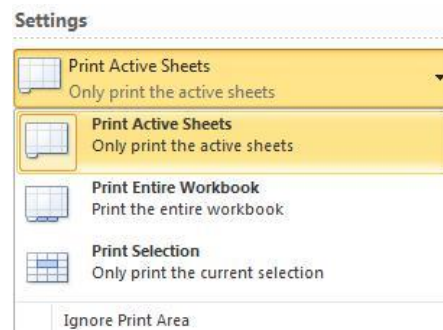
Print Preview automatically displays when you click on the **Print** tab. Whenever you make a change to a print-related setting, the preview is automatically updated.

1. Click the **File** tab, and then click **Print**. To go back to your document, click the **File** tab.
2. A preview of your document automatically appears. To view each page, click the arrows below the preview.

Print a Worksheet

1. Click the worksheet or select the worksheets that you want to print.
2. Click File
3. Click Print.
4. Once you are on the Print screen you can select printing options:

- To change the printer, click the drop-down box under Printer, and select the printer that you want.
- To make page setup changes, including changing page orientation, paper size, and page margins, select the options that you want under Settings.
- To scale the entire worksheet to fit on a single printed page, under Settings, click the option that you want in the scale options drop-down box.



- To print the specific information, select Print Active Sheets or Print Entire Workbook

5. Click Print.

Shortcut Keys Ms-Excel

Action	Keystroke
Open a file	CTRL+O
New file	CTRL+N
Save As	F12
Save	CTRL+S
Print	CTRL+P
Find	CTRL+F
Replace	CTRL+H
Go to	F5
One cell up	up arrow
One cell down	down arrow
One cell right	Tab
One cell left	SHIFT+Tab
Top of worksheet (cell A1)	CTRL+Home
End of worksheet(last cell with data)	CTRL+End
End of row	Home
End of column	CTRL+left arrow
Move to next worksheet	CTRL+PageDown
Formulas	
Apply AutoSum	ALT+=
Current date	CTRL+;
Current time	CTRL+:
Spelling	F7
Help	F1
Macros	ALT+F8
Selecting Cells	
All cells left of current cell	SHIFT+left arrow
All cells right of current cell	SHIFT+right arrow
Entire column	CTRL+Spacebar
Entire row	SHIFT+Spacebar
Entire worksheet	CTRL+A
Text Style	
Bold	CTRL+B
Italics	CTRL+I
Underline	CTRL+U
Strikethrough	CTRL+5

Formatting	
Edit active cell	F2
Format as currency with 2 decimal places	SHIFT+CTRL+\$
Format as percent with no decimal places	SHIFT+CTRL+%
Cut	CTRL+X
Copy	CTRL+C
Paste	CTRL+V
Undo	CTRL+Z
Redo	CTRL+Y
Format cells dialog box	CTRL+1

